Datasheet

Black Oak Engineering

precision electronic instruments

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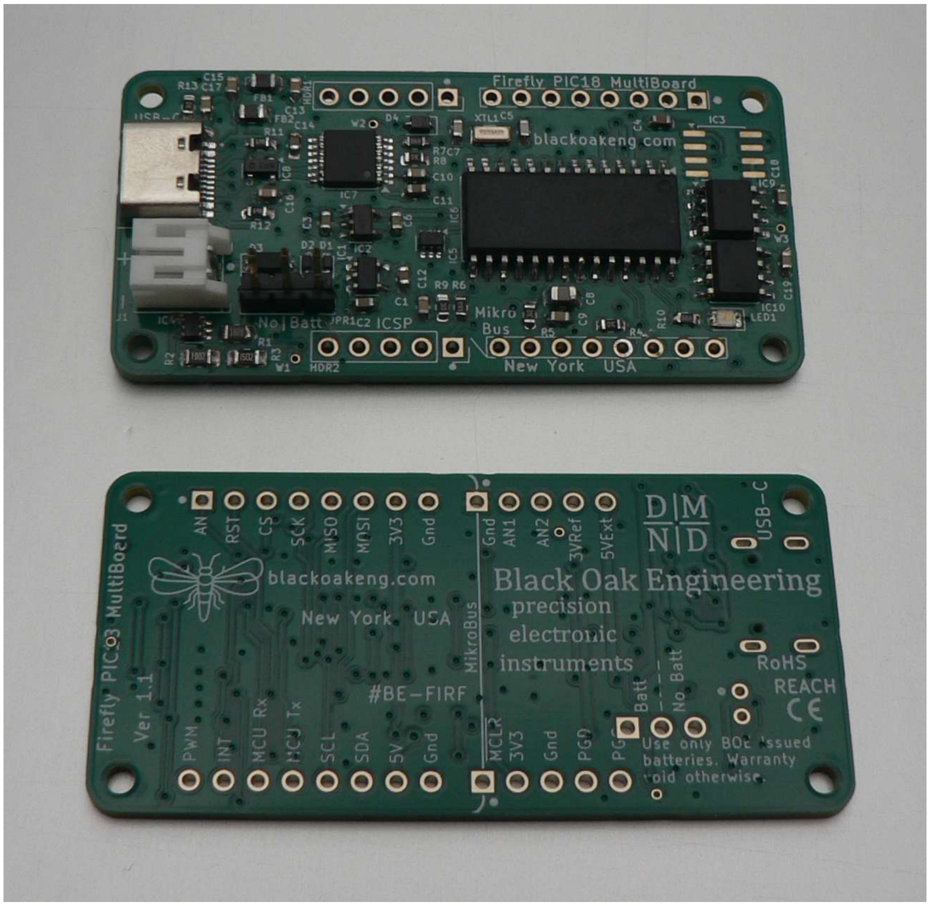
Firefly PIC18 MultiBoard

Part number `BE-FIRF`
Version `1.3`



Description. The Black Oak Engineering (BOE) Firefly PIC18 MultiBoard (#BE-FIRF) goes beyond a typical eval or demo or 'Arduino' board. It is widely used wherever a small, modular, ruggedized, ultra low power board can go. It uses the Microchip PIC18F26Q43 chip. Its breakouts correspond to the popular MikroBus format of which there are hundreds of different types, and there are a few extra breakouts as well. As an open standard, it is easy to design custom mikroBus format boards.

- The Firefly uses USB-C to communicate and for power. That may also recharge the optional lithium battery. With the BOE standard battery, many applications can operate for over a month on one battery charge.

- In addition to the one MikroBus[1] analog input, Firefly adds three more. These are 12-bit precision. An accurate internal voltage reference may be used. One analog pin may be used as an 8-bit DAC.

- An external 5V source may be connected if USB is not available. The direct 'high' voltage from the power source may be used in the breakout (useful for high current loads), or we may use a programmable switched 3.3 V. Thus, all peripheral functions may be unpowered when not in use. The USB section only draws power when USB is connected.

- The USB uses a major PID, so your operating system will recognize it immediately without a problem.

- There is a built in temperature sensor, with ±1°C accuracy, and range −55 to 125°C.

- There is a 64 K x 8 serial EEPROM.

- There is a precise, programmable battery self-test.

- The XLP PIC18 may run off of its internal oscillators (up to 64 MHz). It also has a precise 32.768 kHz quartz crystal, which may be used as a clock source or for accurate time-keeping.

- Firefly uses only parts rated for at least −40 to 85°C. Note, if a battery is used, this will probably have a more limited temperature range. Firefly has several robust EMI countermeasures.

- There are four M2 mounting holes. The design is completely open, so you may easily design your own attaching board. Standard boards exist for analog data acquisition and various sensors. The mikroBus breakout is very general. Most of the pins can be re-tasked to suit your needs. Pins are on 100 mil centers. A variety of header pins or sockets may be attached, top or bottom. There are also breakouts for a QWIIC connector and an extra UART.

- The Firefly is easily programmed, either with MPLAB X or Visual Studio Code. A Board Support Package (BSP) and a complete set of various tested drivers are available at our GitHub. BOE releases source code and other collateral under an attached MIT license.

1 mikroBus is a widely used standard developed by MikroElektronika.

- ICSP breakout for programming. There are several programming tools under $40 available, as well as more expensive, high performance debugging tools.

- Reset tactile switch and programmable LED.

- Designed and manufactured in the USA.

Fireflies form the large family Lampyridae. Most are nocturnal and prefer low, damp ground in temperate regions. Their bioluminescent flashing glow is used to attract mates.

PIC18F26Q43 specifications (partial)

- 8 bit RISC processor.
- 64KB Flash, 4KB SRAM, 1KB internal EEPROM,
- DC – 64 MHz clock. Internal or external.
- DMA, vectored interrupts.
- 25 IO pins with peripheral pin select (remapping).
- 12 bit ADC, 8 bit DAC.
- 2x SPI, 1x I2C, LIN, DALI.
- 4x UART.
- Typical currents. Sleep: 1.2 µA. Full on, 16 MHz, 3.3 V: 2.1 mA.

Firefly PCBA specifications

- 58 × 28 mm (2.2× 1.1 inches).
- 4× 2-56 (or M2) mounting holes.
- Mass = 7.95 g (0.28 oz).

BOE is continuously improving. We also strive to keep one step ahead of procurement shortfalls. We will deliver to you the latest hardware version possible. In some cases specifications will change.

Package includes one 3' (1 m) USB-C cable.

Environmental

- Temperature.   -40 to +85 °C (-40 to +185 °F), excluding battery option.
- Humidity / water exposure. The PCBA does not include a protective enclosure. Nor is it conformally coated. Condensing humidity and water exposure must be completely avoided.

Approvals & Compliance

- RoHS.
- REACH.
- California Prop 65.

Value Added Design. Want to use the Firefly in a new project or OEM application, but need a little assistance? Not a problem. BOE contracts regularly with end users for value added design.

Warranty Policy. Any instrument ordered from BOE may be returned for full refund, less shipping costs, within 30 days of delivery, provided that the instrument has not, in the opinion of BOE been damaged or misused. An RMA number is required in all cases. See our *Standard Terms & Conditions – Instruments* for more details.

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